Problem 26.1

Smith receives \$400 in 1 year, \$800 in 2 years, \$1,200 in 3 years and so on until the final payment of \$4,000. Using an effective interest rate of 6%, determine the present value of these payments at time 0.

Solution

The payments form an increasing annuity-immediate:

$$400, 800, 1200, \ldots, 4000.$$

Since the final payment is \$4,000, the number of payments is

$$n = \frac{4000}{400} = 10.$$

The effective interest rate is

$$i = 0.06,$$

so

$$v = \frac{1}{1.06}.$$

The present value is

$$400(Ia)_{\overline{10}|}.$$

For an increasing annuity-immediate,

$$(Ia)_{\overline{n}|} = \frac{a_{\overline{n}|} - nv^n}{i}.$$

Therefore,

$$PV = 400\left(\frac{a_{\overline{10}|} - 10v^{10}}{0.06}\right).$$

Since

$$a_{\overline{10}|} = \frac{1 - v^{10}}{0.06},$$

we have

$$PV = 400\left(\frac{\frac{1-(1.06)^{-10}}{0.06} - 10(1.06)^{-10}}{0.06}\right).$$

Thus,

$$PV = 14784.96337.$$

Therefore, the present value is

$$\boxed{14784.96}.$$